

Wireless Baby Monitor

JOY MUKHERJI

A baby monitor can help you find peace of mind. You can now monitor your sleeping baby with this wireless baby monitor. It is a radio system used to remotely listen to sounds made by an infant. Simply place the circuit near your sleeping baby and listen through an FM receiver from any other room in the house. It can also alert you if the baby wakes up in the middle of the night.

Circuit and working

The circuit is built around a low-power audio amplifier using LM386 (IC1), hex inverting Schmitt trigger 74HC14 (IC2), voltage regulator 7805 (IC3), 10MHz crystal (XTAL1), varactor diode 1SV149 (D1) and a few other components. A parallel resonant oscillator circuit is formed around inverter gate N1 along with crystal (XTAL1), resistor R3, capacitors C3 and C4 and varactor diode

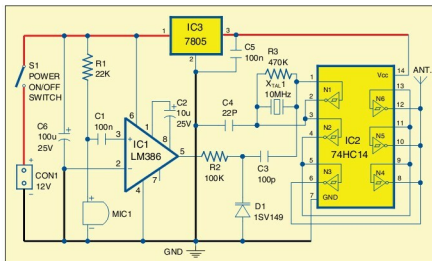


Fig. 1: Circuit diagram of the wireless baby monitor

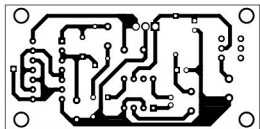


Fig. 2: PCB of the baby monitor circuit

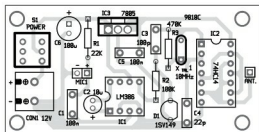


Fig. 3: Component layout of the PCB

D1. It generates square waves at the fundamental frequency of 10MHz of crystal.

The signal is buffered by gate N2 and further boosted by parallel inverter gates N3, N4, N5 and N6. Unlike sine waves, square waves have many harmonics above their fundamental frequency. The monitor transmits on tenth (100MHz) harmonics of the square wave. Use a quarter-wave 75cm piece of wire for the antenna.

The audio section of the transmitter is built around IC1. The gain is set to 200 by capacitor C2. Audio from electret



PARTS LIST

Semiconductors:

- IC1 - LM386 low-power audio amplifier
- IC2 - 74HC14 hex inverting Schmitt trigger
- IC3 - 7805, 5V voltage regulator
- D1 - 1SV149 varactor diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 22-kilo-ohm
- R2 - 100-kilo-ohm
- R3 - 470-kilo-ohm

Capacitors:

- C1, C5 - 100nF ceramic disk
- C2 - 10 μ F, 25V electrolytic
- C3 - 100pF ceramic disk
- C4 - 22pF ceramic disk
- C6 - 100 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- S1 - On/off switch
- MIC1 - Electret microphone
- XTAL1 - 10MHz crystal
- ANT.1 - 75cm long wire
- 12V adapter

microphone MIC1 is picked up and amplified by LM386. It is then coupled with varactor diode D1 via resistor R2. The crystal's frequency along with D1 forms an FM modulation signal. Since the circuit transmits on tenth harmonic of crystal frequency, audio deviation is also multiplied by a factor of 10. This results in clear audio that can be received on an FM receiver.

Construction and testing

An actual-size, single-side PCB for the baby monitor circuit is shown in Fig. 2 and its component layout in Fig. 3. Keep all leads as short as possible. After assembling the circuit on a PCB, enclose it in a suitable plastic box. Drill a small hole for the microphone. Use 12V to power the circuit. The current consumption of the circuit is very low. Before using the circuit, ensure that power supply is correct. ●



The author is an electronics hobbyist and a small-business owner in Albany, New York, USA. His interests include designing radio frequency circuits